

Abhinav Karnatak

Full-Stack Developer | Applied AI & Agentic Engineering

abhinavkarnatak2004@gmail.com • +91-7310988194 • [linkedin.com/in/abhinavkarnatak](https://www.linkedin.com/in/abhinavkarnatak)
github.com/abhinavkarnatak-dev • abhinavkarnatak.com

Full-stack developer building **AI-powered applications** and **agentic systems** using **LangGraph, LangChain, RAG, and LLM integrations**. Hands-on experience building end-to-end systems with **Typescript, Python, Node.js, and cloud infrastructure**, with a focus on **agent workflows, tool execution, evaluation, and reliable software systems**.

SKILLS

AI/LLM Engineering: LangGraph, LangChain, Agentic Workflows, Tool Calling, LLM Evaluation, Guardrails, Prompt Engineering, RAG, LLM API Integration

Programming Languages: Python, TypeScript, JavaScript, C++

Backend & Integrations: Node.js, Express.js, REST APIs, OAuth 2.0, JWT, WebSockets, Socket.io, RabbitMQ, Redis

Databases: MongoDB, PostgreSQL, Prisma

Cloud & DevOps: Docker, AWS (EC2, S3), E2B, Nginx

Frontend Technologies: React.js, Next.js, React Native, Tailwind CSS, Framer Motion, Progressive Web Apps (PWA)

Libraries & Integrations: Gemini API, Razorpay

PROJECTS

Nimbus: A Cloud Coding Agent 🔗

- Built a **cloud coding agent** using **TypeScript, Node.js, React, MongoDB, Redis, and LangGraph** that connects to **GitHub repositories**, understands and navigates codebases, answers repository-level queries, and autonomously works on coding tasks.
- Engineered an **agentic workflow with LangGraph** for iterative **LLM reasoning, codebase retrieval, and tool execution**, enabling the agent to inspect files, modify code, execute commands, and validate changes through **build, lint, and test checks**.
- Designed a secure **E2B sandbox architecture** where model-generated commands and code changes execute in an **isolated environment** without sensitive credentials, while the trusted backend validates resulting patches and handles authenticated **GitHub operations and pull requests**.

YapIt: Real-Time Microservices Chat Application 🔗

- Built a **real-time** full stack application using **Next.js, Node.js, Socket.io, and Redis**, supporting **persistent WebSocket-based communication** between users.
- Engineered a decoupled **Microservices Architecture** (User, Chat, and Mail services) utilizing **RabbitMQ** for reliable inter-service messaging and asynchronous task handling like global account data cleanup.
- Containerized the entire ecosystem with **Docker Compose** and deployed it to an **AWS EC2** instance, utilizing **Nginx** reverse proxy to securely manage API routing, continuous WebSocket traffic, and AWS S3 file-sharing.

Revuea: Empowering Honest Feedback with Anonymity 🔗

- Developed a full-stack web application using **React.js, Node.js, and PostgreSQL** (via **Prisma ORM**) to securely collect, store, and manage anonymous user feedback.
- Implemented a secure authentication flow using **OTP-based signups** and built a dashboard utilizing **Recharts** to visualize feedback data efficiently.
- Integrated the **Gemini AI API** to instantly process large volumes of user feedback into readable text summaries, while also adding **CSV export** for external data analysis.

PixScribe: SaaS Text-to-Image Generator Web App 🔗

- Built using **React.js, Node.js, Express, and MongoDB**, PixScribe is a full-stack web app that lets users generate AI-based images from text prompts.
- Engineered a credit system linked to a **Razorpay** integration, handling secure payment processing and automatically updating user account balances upon successful transactions.
- Designed a fast, highly responsive user interface with **Tailwind CSS** and integrated **Framer Motion** to deliver smooth layout animations and state transitions.

WORK EXPERIENCE

Analyst (Software Engineer)

Sep 2025 - May 2026

Capgemini

Mumbai, Maharashtra

- Assisted in troubleshooting network and connectivity issues, gaining hands-on exposure to enterprise server environments (Linux/Windows) while ensuring timely resolution of assigned tickets.
- Collaborated with the team to monitor system health and ensure smooth daily operations.

Data Engineer Intern

Jun 2024 - Jul 2024

HERE Technologies

Remote

- Built custom Python web crawlers using **Scrapy** to scrape store locations from retail websites with varying HTML layouts, scheduling automated runs that cut manual effort by 30%.
- Documented the workflow and codebase to make it easier to maintain and update in the future.

EDUCATION

B.Tech Computer Science and Engineering

Aug 2021 - May 2025

University Of Petroleum and Energy Studies

- Served as the **Student Placement Representative (SPR)**, coordinating communication between the placement cell and students during campus recruitment activities.